



DPUOscillator

Signal generator.

1. Inputs:

<i>In</i>	Input signal
<i>On</i>	On/Off switch of the signal
<i>Start</i>	Start impulse of oscillator

2. Dynamic parameters:

<i>Period</i>	Period of signal in milliseconds. Default 100 ms
<i>Amplitude</i>	Amplitude of signal. Default : 1
<i>Offset</i>	Offset of signal. Default: 0
<i>PulsePause</i>	Puls/Pause ratio of impulse signal. Default: 0.5
<i>PhaseLag</i>	Phase delay of signal in milliseconds. Default: 0 ms
<i>Signal</i>	Signal type (sine, triangle, rectangle, impulse). Default: 0 (Sine)

3. Outputs:

<i>Out</i>	Output of signal
------------	------------------

4. Functional principle:

DPUOscillator represents a signal generator, which adds a periodic signal to the input signal.

$$Out = \begin{cases} In + Signal(t) & | \quad On = 1 \\ In & | \quad On = 0 \end{cases}$$

As signal wave form the following signal types can be set with the parameter *Signal*.

Signal	Signal type
0	Sine
1	Triangle
2	Rectangle
3	Rectangle impulse
4	Ramped
>4	No signal

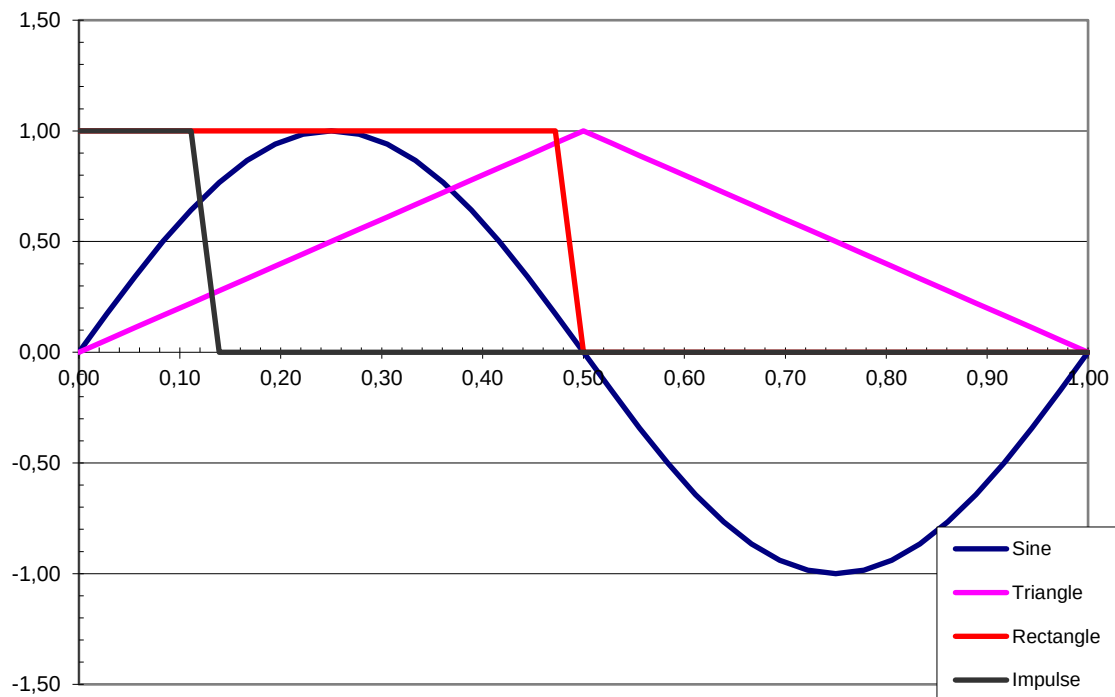


The oscillator must be started once via the input *Start*. This start impulse defines the starting time point (internal parameter *StartTime*) of the period. This is required to synchronize different signal generators with each other.

$$\text{Signal}(t) = (\text{Generator}((t - \text{StartTime}) + \text{PhaseLag}, \text{Period}) \cdot \text{Amplitude}) + \text{Offset}$$

While the input *On* is 1, the generated signal is added to the input signal. The inputs *Period*, *Amplitude*, *PhaseLag* and *Offset* can be changed online. Thus, by interconnecting several generators, e.g. the period of one generator can be controlled by the output of another generator.

The signal wave forms are displayed in the following figure:



5. Example:

The following is an example script for using DPUOscillator:

```
DPUKeyboardState keyboard
{
    Computer = {LOCALHOST};
    Index = 1;
};

DPUOscillator oscillator
{
    Computer = {LOCALHOST};
    Index = 1;
```



SILAB DRIVING SIMULATION

version Version 7.1 documentation

```
In = -1;
On = 1;
Amplitude= 2;
Period = 314;
Signal = 2;

};
Connections =
{
    keyboard.S -> oscillator.Start
};
```

In the given example, a rectangle wave (signal=2) is generated. Its amplitude is 2 and its period is 314 ms. When “S” is pressed on the keyboard, the wave starts oscillating.